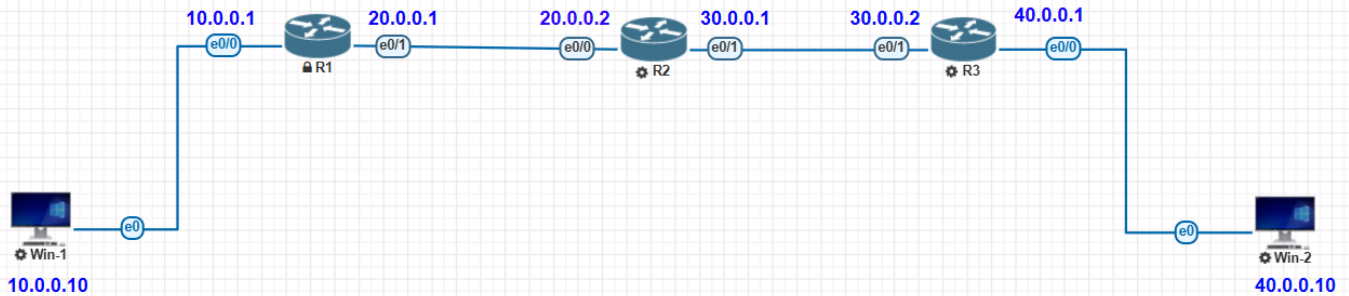


EIGRP CONFIGURATION



Step 1: Assign ip address in all cisco routers

R1					
R1#sh ip int br					
Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	10.0.0.1	YES	TFTP	up	up
Ethernet0/1	20.0.0.1	YES	TFTP	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R2					
R2#sh ip int br					
Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	20.0.0.2	YES	TFTP	up	up
Ethernet0/1	30.0.0.1	YES	TFTP	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R3					
R3#sh ip int br					
Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	40.0.0.1	YES	TFTP	up	up
Ethernet0/1	30.0.0.2	YES	TFTP	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

Step 2 : Check interface parameter

```
R1#sh int eth0/0
Ethernet0/0 is up, line protocol is up
  Hardware is AmdP2, address is aabb.cc00.0100 (bia aabb.cc00.0100)
  Internet address is 10.0.0.1/8
  MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Auto-duplex, Auto Speed, media type is unknown media type
  input flow-control is off, output flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
```

Step 3 : Enable eigrp protocol in all cisco router

```
R1
R1(config)#router eigrp 1
R1(config-router)#
R1(config-router)#network 10.0.0.0
R1(config-router)#network 20.0.0.0
R1(config-router)#exit
R1(config)#

R2
R2(config)#router eigrp 1
R2(config-router)#
R2(config-router)#network 20.0.0.0
R2(config-router)#net
*Sep 22 03:57:03.721: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor 20.0.0.1 (Ethernet0/0) is up: new adjacency
R2(config-router)#network 30.0.0.0
R2(config-router)#exit
R2(config)#

R3
R3(config)#
R3(config)#router eigrp 1
R3(config-router)#network 30.0.0.0
R3(config-router)#netw
*Sep 22 03:57:40.211: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor 30.0.0.1 (Ethernet0/1) is up: new adjacency
R3(config-router)#network 40.0.0.0
R3(config-router)#exit
R3(config)#
```

Step 4 :Check eigrp neighbor table

```
R1
R1(config)#exit
R1#s
*Sep 22 03:59:16.282: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H   Address                Interface          Hold Uptime    SRTT    RTO    Q
Seq                                (sec)          (ms)          Cnt
Num
0   20.0.0.2                Et0/1             12 00:02:36    5    100    0
8
R1#
```

```
R2
R2(config)#do sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H   Address                Interface          Hold Uptime    SRTT    RTO    Q
Seq                                (sec)          (ms)          Cnt
Num
1   30.0.0.2                Et0/1             13 00:02:35    5    100    0
4
0   20.0.0.1                Et0/0             13 00:03:12    4    100    0
4
R2(config)#
```

```
R3
R3(config)#do sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H   Address                Interface          Hold Uptime    SRTT    RTO    Q
Seq                                (sec)          (ms)          Cnt
Num
0   30.0.0.1                Et0/1             13 00:02:53    4    100    0
7
R3(config)#
```

Step 5 : Check eigrp topology table

R1

```
R1#sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(1)/ID(20.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 10.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/0
P 30.0.0.0/8, 1 successors, FD is 307200
   via 20.0.0.2 (307200/281600), Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 332800
   via 20.0.0.2 (332800/307200), Ethernet0/1
P 20.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/1
```

R2

```
R2(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(1)/ID(30.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 10.0.0.0/8, 1 successors, FD is 307200
   via 20.0.0.1 (307200/281600), Ethernet0/0
P 30.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 307200
   via 30.0.0.2 (307200/281600), Ethernet0/1
P 20.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/0
```

R3

```
R3(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(1)/ID(40.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 10.0.0.0/8, 1 successors, FD is 332800
   via 30.0.0.1 (332800/307200), Ethernet0/1
P 30.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 281600
   via Connected, Ethernet0/0
P 20.0.0.0/8, 1 successors, FD is 307200
   via 30.0.0.1 (307200/281600), Ethernet0/1
```

Step 6 : Check routing table

